RADOME

Document Version 1.1 / Architecture Revision 3

Distributed Multiband Conformal Radome Network

José Augusto M de Andrade Jr.

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Abstract

This project defines a distributed passive electromagnetic sensing network based on geodesic and conformal radomes. It integrates multiband vector reception, opportunistic illumination, local event processing, precise timing, end-to-end calibration, passive multistatic localization and resilient mountain infrastructure. The document is conceptual and intended for research, simulation, prototyping and experimental validation.

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Chapter 1

Scope, Motivation and Project Status

The RADOME project proposes a distributed passive electromagnetic sensing network based on geodesic and conformal radomes installed at elevated sites. Each station combines multiband receiving apertures, local digitization, event-oriented processing, precise timing, calibration and optical networking. Several stations cooperate to estimate angle of arrival (AOA), time difference of arrival (TDOA), frequency difference of arrival (FDOA), Doppler and polarization.

The term *radome* derives from *radar dome*: an enclosure placed over an antenna to protect it from wind, temperature, dust, rain and snow while seeking electromagnetic transparency [16]. In this project, however, the dome does not house a dedicated radar transmitter. It protects and integrates receiving apertures only. The radar function emerges from cooperation among stations: each node observes direct emitter signals or signals from external illuminators reflected by targets, and the network estimates position and motion through AOA triangulation and TDOA/FDOA/Doppler multilateration and fusion. The concept is therefore a passive multistatic radar built from receiving radomes, not a set of distributed active radars.

Receive-only operation makes the station substantially less observable in the emission domain than an active radar: it generates no illumination pulses, RF power-amplifier output or scanning transmission that discloses its position and operating mode. The system listens to external waveforms of known origin and geometry; any RF energy attributable to its presence by an external sensor is secondary scattering—an echo of ambient illumination—rather than an interrogation emission generated by the radome. This electromagnetic discretion is an architectural property. It is distinct from the structure's radar cross section, whose magnitude depends on materials and geometry.

Many conventional radomes are developed for active radars, for which the enclosure must preserve both the transmitted wave and the received echo. Wide coverage, scan angle and polarization require co-design of permittivity, layer thicknesses, cores and multilayer stacks, together with specialized electromagnetic manufacturing and validation [16, 17]. RADOME investigates a different cost–performance balance: the receiving Yagis are exposed and oriented by each face normal, so their useful path does not depend on multiband shell transparency. This may simplify module-level characterization and replacement, but transfers requirements to exposed supports, sealing, cables, corrosion, wind, lightning and recalibration. Lower acquisition or life-cycle cost is therefore a hypothesis to be quantified, not a demonstrated result.

The network may exploit known opportunistic illuminators, including cellular infrastructure, broadcast transmitters, satellites and other authorized RF sources. In bistatic operation, the direct reference path is compared with the path involving a target and a receiver. The system is conceptual: performance, coverage, locations and accuracy must be demonstrated experimentally and are not operational guarantees.

The project objectives are to develop a measurable multiband architecture, validate a three-node VHF/UHF demonstrator, characterize end-to-end timing and RF delays, preserve event windows locally, and integrate the electromagnetic platform with resilient energy, thermal, communications and site infrastructure.

The independent English edition, assembled from `radome-en.tex`, is one of the two authoritative technical publications. “Document version 1.1” identifies the controlled publication, while “architecture revision 3” identifies the current conceptual system architecture; these identifiers are independent. Earlier Markdown, PDF and standalone LaTeX documents are historical inputs unless a current chapter explicitly incorporates their content. Quantitative values are controlled in `PARAMETERS.md`, with status proposed, simulated, measured or derived, and architecture decisions are recorded in `DECISIONS.md`. In case of conflict, an approved parameter-register entry and this article take precedence over historical material.

1.0.1 Authorship and editorial control

José Augusto M de Andrade Jr. is the sole author of this article. The selection and formulation of ideas, organization of content, approved wording and compilation instructions belong exclusively to the author. Bibliographic sources, computer programs and tools assisting research, writing, translation, illustration or compilation are supporting means and do not constitute co-authorship. No contribution produced with these means enters an authoritative edition without the author’s direction and final approval.

Chapter 2

System Architecture

The network is hierarchical. Local clusters with baselines of tens of kilometres serve regional illuminators and low-altitude scenarios. Regional clusters extend coverage, while long baselines may support HF, high-altitude and space-object observations when common visibility and a compatible illuminator exist.

Each station contains a radome platform, multiband RF chains, local processing, timing and calibration, optical transport, power conditioning, thermal management and a supervisory interface. The central fusion layer associates detections by waveform, time, frequency, Doppler, polarization and geometric consistency. It must distinguish direct emitter observations, bistatic reflections, interference and tracking hypotheses.

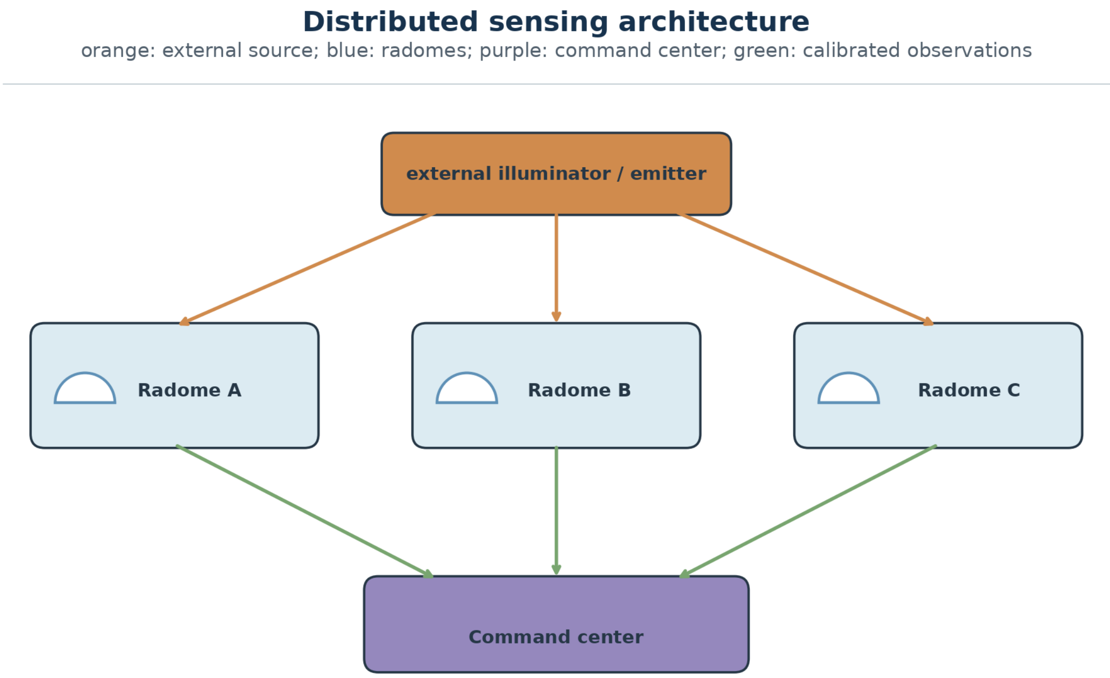

Figure 2.1: Distributed sensing architecture: the upper block is an external illuminator or emitter, the three blue blocks identify Radomes A, B and C, green paths carry calibrated observations, and the lower purple block is the command center responsible for association and fusion.

Chapter 3

Geodesic Geometry and Radome Structure

Architecture revision 3 starts from a proposed regular icosahedron (parameters `GEO-001` and `GEO-002`) with $V_0 = 12$, $E_0 = 30$ and $F_0 = 20$. The tetrahedral candidate in ADR-012 uses a class-I, frequency-2 planar subdivision: each equilateral macroface is split at its edge midpoints into four coplanar equilateral receiver faces, without projecting the new points to a sphere. The closed envelope verified by `geometry/verify_tetrahedral_face_geometry.py` remains

$$V = 42, \qquad E = 120, \qquad F = 80,$$

with Euler consistency $V - E + F = 2$ (`GEO-003`, `GEO-009` and `GEO-010`). Every external receiver face now has three 2.0000 m edges (`GEO-004` and `GEO-015`); each macroface edge is 4.0000 m. The resulting faceted envelope has circumradius 3.8042 m and inradius 3.0230 m (`GEO-017`). These values replace the projected-sphere dimensions for this candidate.

A literal regular tetrahedron with all six edges equal to 2 m cannot be repeated over this closed inward assembly: the verifier detects 120 volumetric interpenetrations (`GEO-019`). The adopted interpretation keeps only the three external edges at 2 m and uses an independent inward apex 0.75 m behind each face centre (`GEO-021`). The 80 shallow tetrahedral cells have no volumetric collision (`GEO-020`), and their lateral edges are 1.3769 m. Adjacent apexes are separated by 0.5435 m to 1.1547 m; the two independent Faraday walls leaving each shared external edge therefore bound an inter-cell corridor for separated power, optical fibre, bonding and maintenance routes (`MEC-003` and `MEC-005`). The construction reserves an internal free-core radius of at least 2.2730 m (`GEO-022`). The former circular cut and support ring belong to the projected candidate and are historical; C2 remains open until a boundary made of complete modules and a new civil transition are verified.

The mechanical skeleton, dielectric panels and internal RF modules have different but coordinated geometries. A face is a mechanical and computational sector, not an electromagnetically isolated region. Angular overlap between neighbouring faces is required for calibration, interpolation and direction estimation.

The external RF aperture occupies the unshielded triangular face. For any band declared dual-polarized, two simultaneous orthogonal antenna components follow the local tangent basis $\mathbf{t}_u, \mathbf{t}_v$ and use independent feeds and shielded ADC/ASIC enclosures; coherence and polarization products remain subject to the same-band calibration requirements of C1. The other three tetrahedral faces are conductive Faraday walls. Penetrations use filtered feedthroughs, and the enclosures bond to a conductive cold plate without making the dielectric aperture the primary structural load path.

The internal joints form continuous load paths between the triangular frame members and the neighbouring faces. Sealed structural seams, local gussets and an internal ring or node plate transfer wind, vibration and maintenance loads without relying on the dielectric skin alone. The joint design must be verified by structural analysis and environmental tests.

3.1 Structural resolution

The external Yagi load is transferred through the base mast into the triangular frame and then into the internal node and neighbouring face joints. The dielectric skin is an RF and environmental enclosure, not the sole primary load path. The structural model must include wind pressure on the Yagi and radome, mast bending, equipment mass, thermal expansion, vibration, maintenance loads and lightning-current paths. This closes the architectural gap between an antenna concept and a supportable face module, while finite-element analysis, modal testing and climatic qualification remain required.

The Blender model makes the mounting convention explicit (`MEC-006` and `MEC-007`): the triangular RF face lies on the outer radome surface, while the pyramid volume and its apex are directed inward. A support strut starts at the internal apex, crosses the centre of the face and continues outward as the common boom axis, collinear with the external normal of that face. The VHF and UHF elements are transverse to the normal, mutually orthogonal, and the complete crossed assembly is rotated by 45 degrees around the normal relative to the local tangent basis. This mechanical rotation does not make the different-band channels a coherent polarimetric pair.

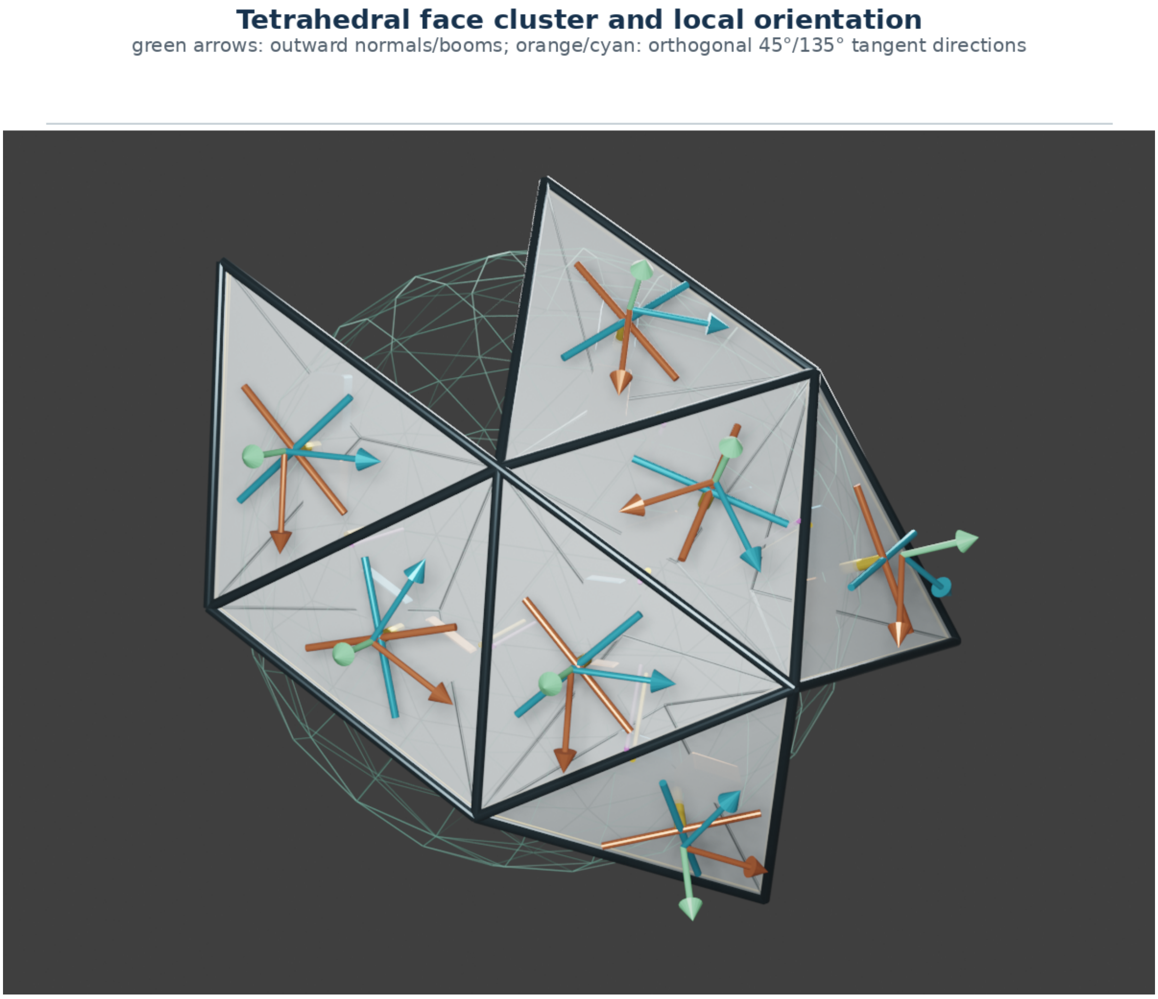

Figure 3.1: Seven contiguous shallow tetrahedral cells. Orange and cyan members show the mutually orthogonal VHF/UHF directions at 45° and 135° in each local tangent basis; green arrows show the outward face normals, which define the boom directions. The green wire envelope is the reserved internal service core.

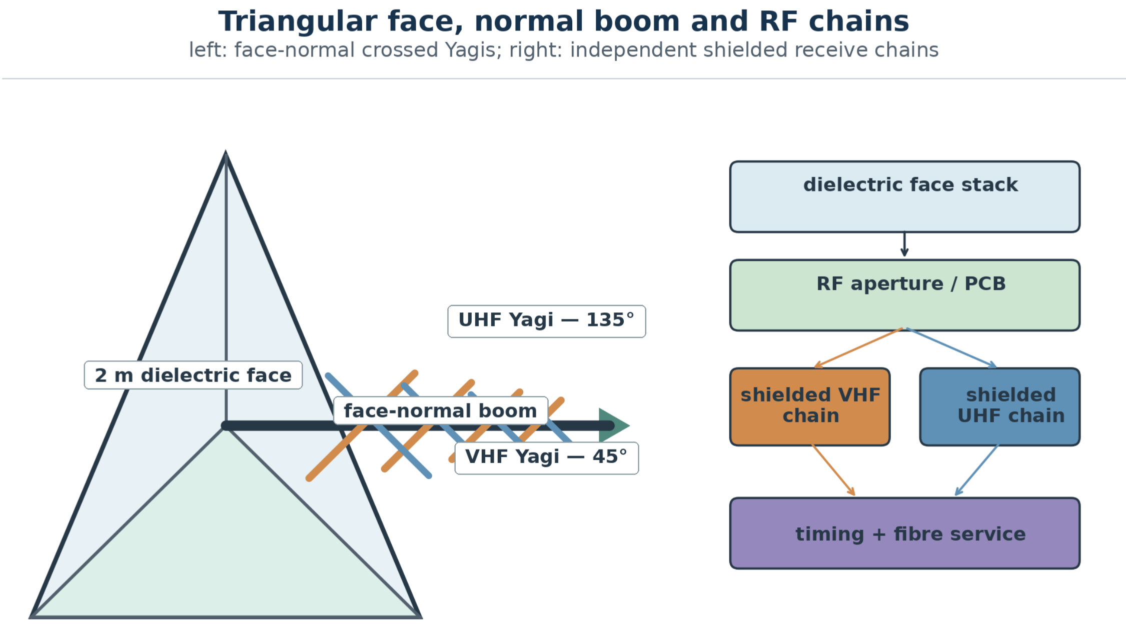

Figure 3.2: Face assembly. The dark boom follows the outward normal from the inward apex through the centre of the two-metre triangular aperture; orange and blue elements are mutually orthogonal and jointly rotated 45° around that normal. At right, the aperture feeds independent shielded VHF and UHF chains connected to the common timing and optical-fibre service.

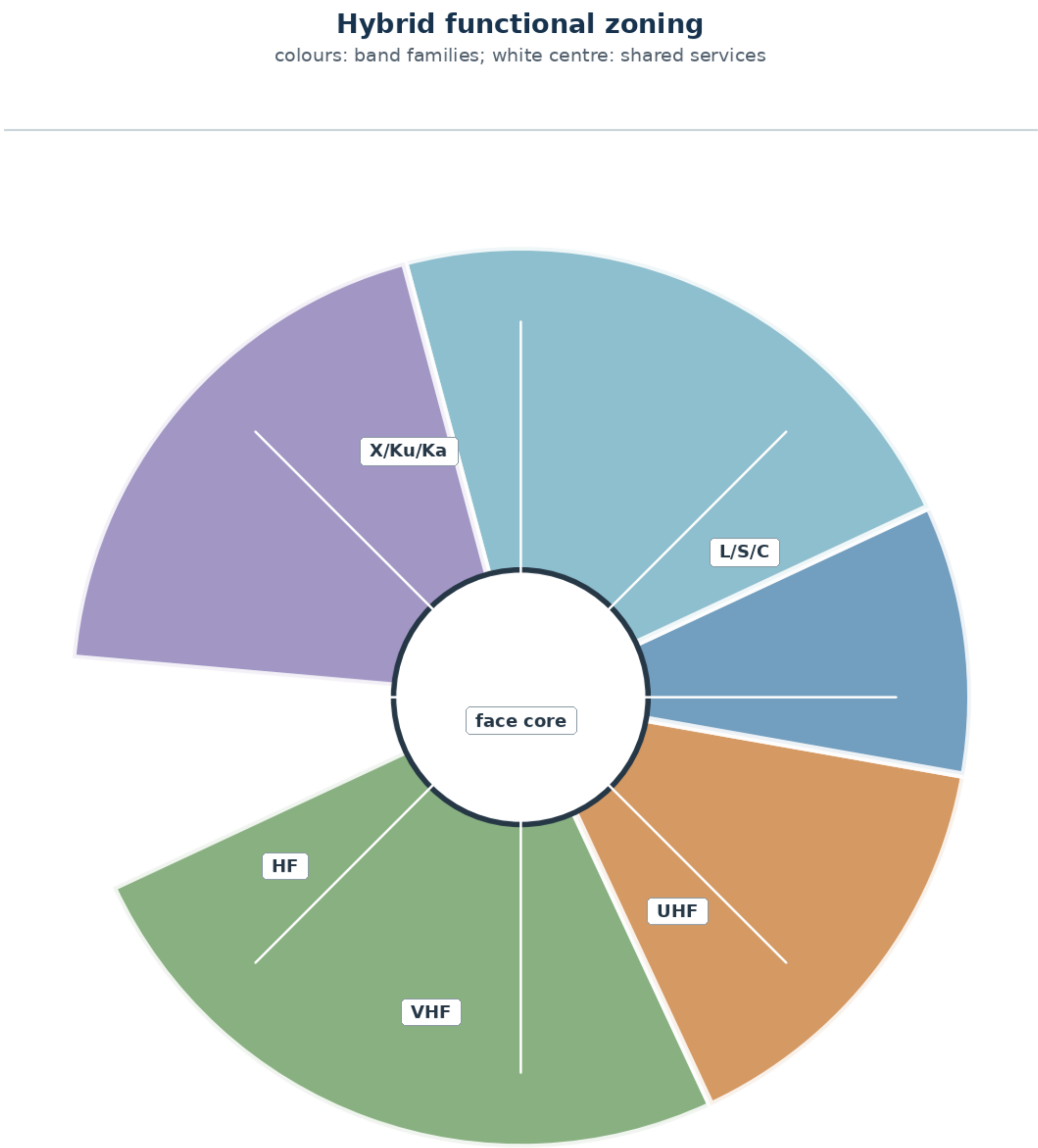

Figure 3.3: Functional zoning: coloured sectors denote the hybrid band families and the white centre denotes shared face-level timing, calibration, processing and services. Sector boundaries are architectural allocations, not rigid electromagnetic boundaries.

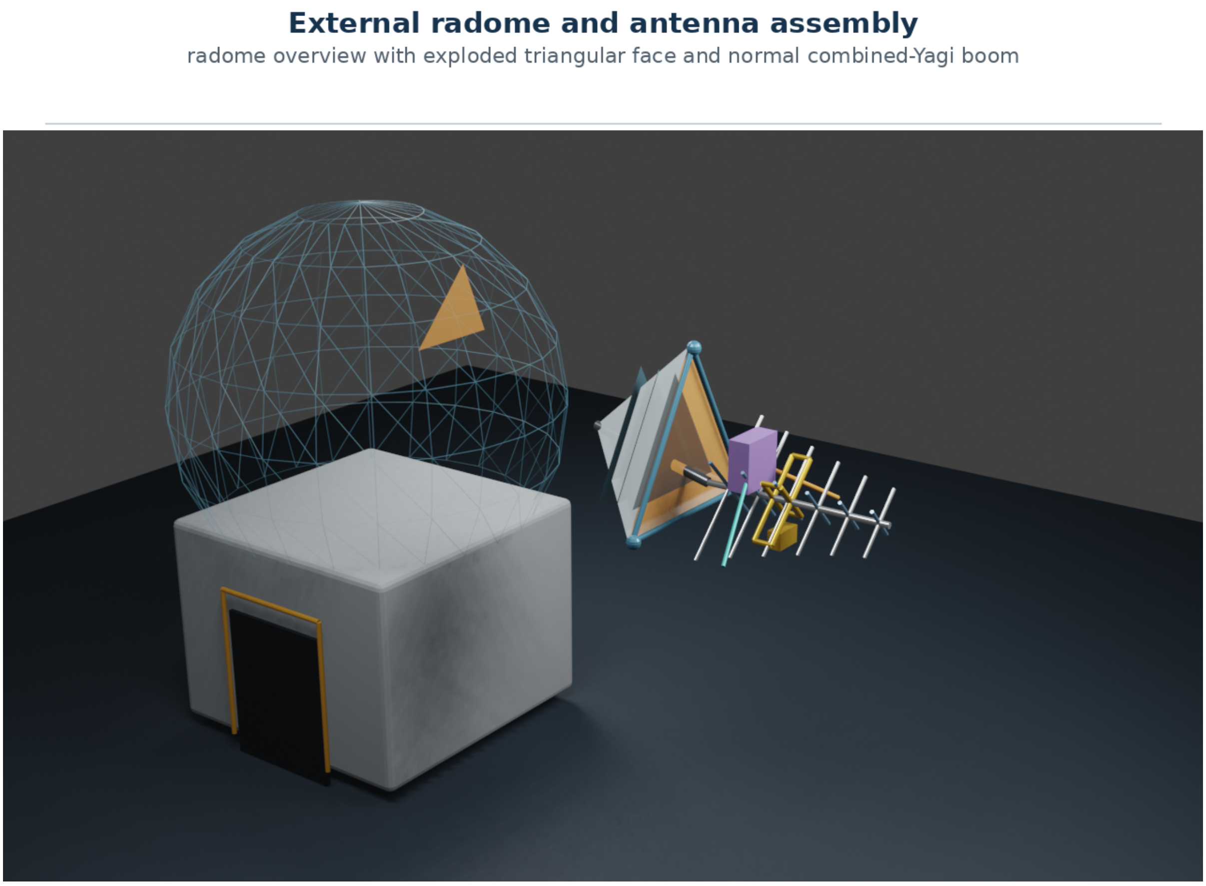

Figure 3.4: External Blender view of the radome and exploded face. The combined VHF/UHF boom is collinear with the face normal; the transverse elements are orthogonal and rotated 45° around that normal.

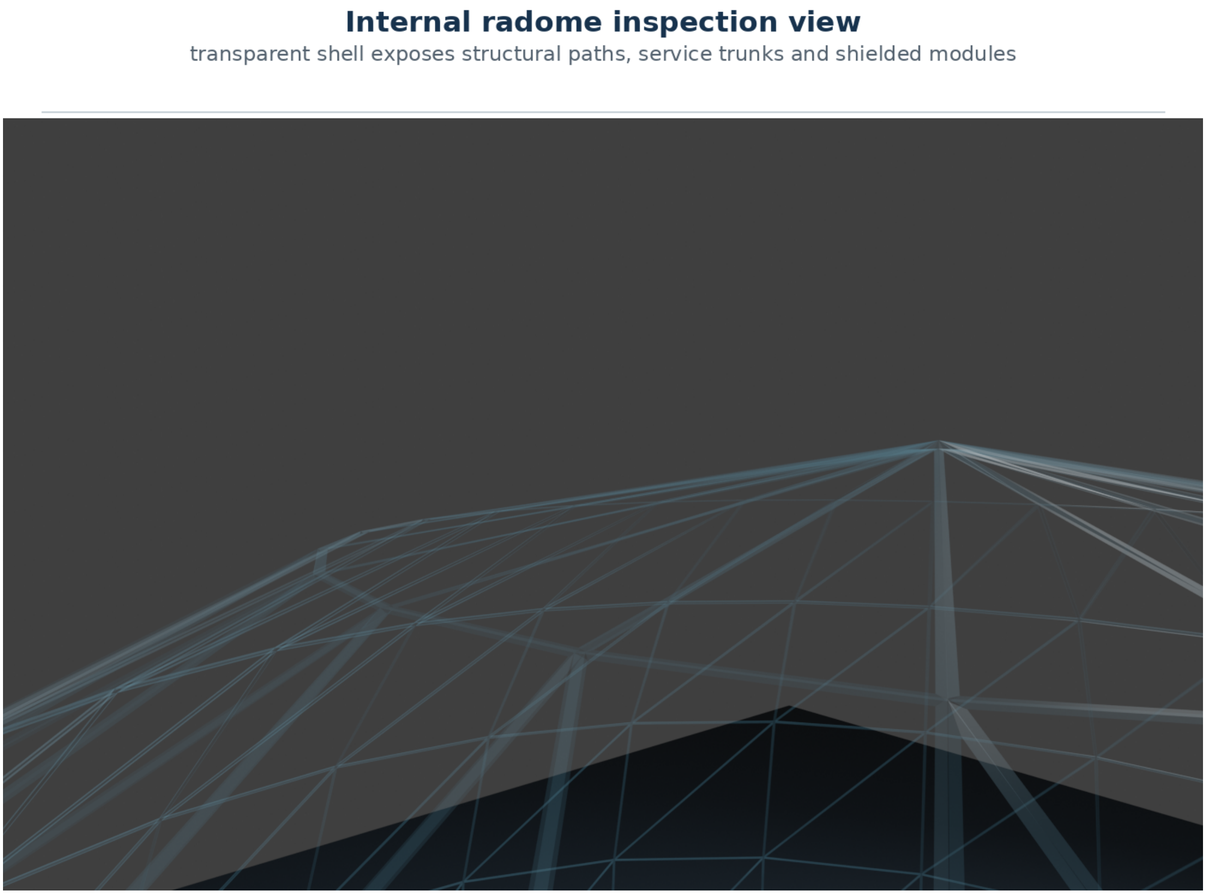

Figure 3.5: Internal Blender inspection view showing the transparent shell, structural load paths, service trunks and representative shielded modules.

Chapter 4

Multiband Aperture and Polarimetry

The aperture is intentionally hybrid rather than uniform across the spectrum. One external single-polarization VHF Yagi and one external single-polarization UHF Yagi share a boom aligned with the outward face normal: the VHF antenna uses larger elements and the UHF antenna uses smaller elements. Their transverse directions are mutually orthogonal at 45° and 135° in the local tangent basis. These mechanical orientations provide diversity across different bands, not two polarization components of one signal. Smaller L/S/C, X/Ku and K/Ka antennas or tiles occupy the face layers and local mounts. HF uses loops, crossed dipoles and platform-mode diversity. Each antenna family has its own filtering, conversion and sampling strategy, while the face preserves a common timing, calibration and fusion interface.

The first demonstrator does not claim continuous spectral coverage. The 323–470 MHz and 860–960 MHz intervals are deliberately unassigned. A dedicated 960–1215 MHz aviation aperture or array supplies two simultaneous independent paths: a 978 MHz UAT preselector, limiter/LNA, coherent ADC and FPGA channel; and a 1090 MHz 1090ES preselector, limiter/LNA, coherent ADC and FPGA channel. Neither path is assigned to the 470–860 MHz UHF Yagi, and the aviation aperture does not imply continuous monitoring across its full tuning range.

Polarimetric synthesis is permitted only in a same-band module that provides two simultaneous, coherent orthogonal ports at the same frequency (`POL-001`). In a local x–y basis:

$$E_{\mathrm{RHCP}} = \frac{E_x - jE_y}{\sqrt{2}}, \qquad E_{\mathrm{LHCP}} = \frac{E_x + jE_y}{\sqrt{2}}.$$

The equations do not combine the VHF and UHF Yagis. The first VHF/UHF demonstrator treats them as independent single-polarization channels (`POL-002`); full polarimetry requires a later dual-port or same-band crossed module. For a curved surface, valid local bases must be rotated to a common coordinate system before fusion. Polarimetric calibration is a function of frequency, angle, temperature, amplitude balance, phase and mutual coupling.

A separate future four-channel mast, distinct from the illustrated combined VHF/UHF pair, implements VHF-X/VHF-Y at 0°/90° and UHF-A/UHF-B at 45°/135°, with four electrically independent booms and RF chains (`POL-004`–`POL-006`). For each band b, calibration follows:

$$\widehat{\mathbf{E}}_b(f,\theta,\phi,T) = \mathbf{C}_b^{-1}(f,\theta,\phi,T)\,[\mathbf{v}_b - \widehat{\mathbf{n}}_b]\,,$$

where $\mathbf{C}_b$ is the measured complex 2×2 Jones matrix containing embedded patterns, gain/phase imbalance, delays, mutual coupling, mast and platform effects. A regularized pseudoinverse replaces the direct inverse where conditioning is poor. The UHF estimate is then rotated by $\psi = 45°$ into the common global basis before polarization products are formed.

4.1 Engineering solution

The crossed-band Yagi assembly is a proposed mechanical integration solution. The larger Yagi covers VHF and the smaller Yagi covers UHF; each remains independently matched, digitized and calibrated. Their outputs may be compared or fused as different-band observations, but no relative phase operation produces RHCP, LHCP, Jones or Stokes quantities from this pair. Novelty and performance are not claimed until prior-art review and measurement.

For wider VHF discovery coverage, an 11-element 72–320 MHz LPDA with a 2 m maximum element is retained as a simulation candidate (`ANT-001–ANT-005`). It is not yet a replacement for the illustrated VHF Yagi: the approximately 2 m boom, face-edge clearance, wind load, feed transition, gain and interaction with the UHF antenna require validation.

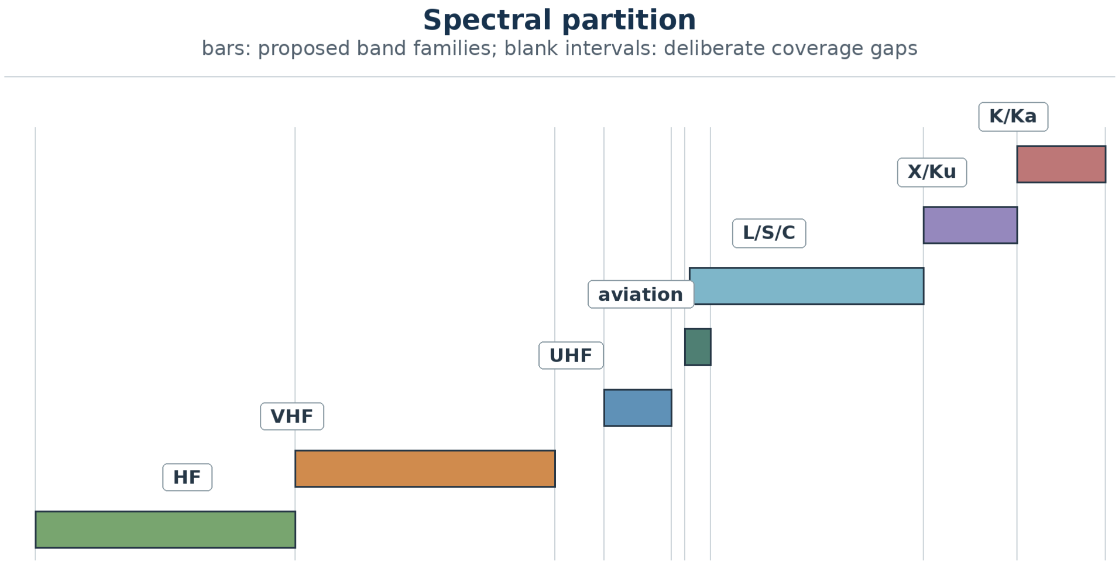

Figure 4.1: Spectral partition on a logarithmic horizontal scale. Each coloured bar is one proposed band family; the blank intervals between the VHF, UHF and dedicated aviation paths are deliberate first-demonstrator gaps.

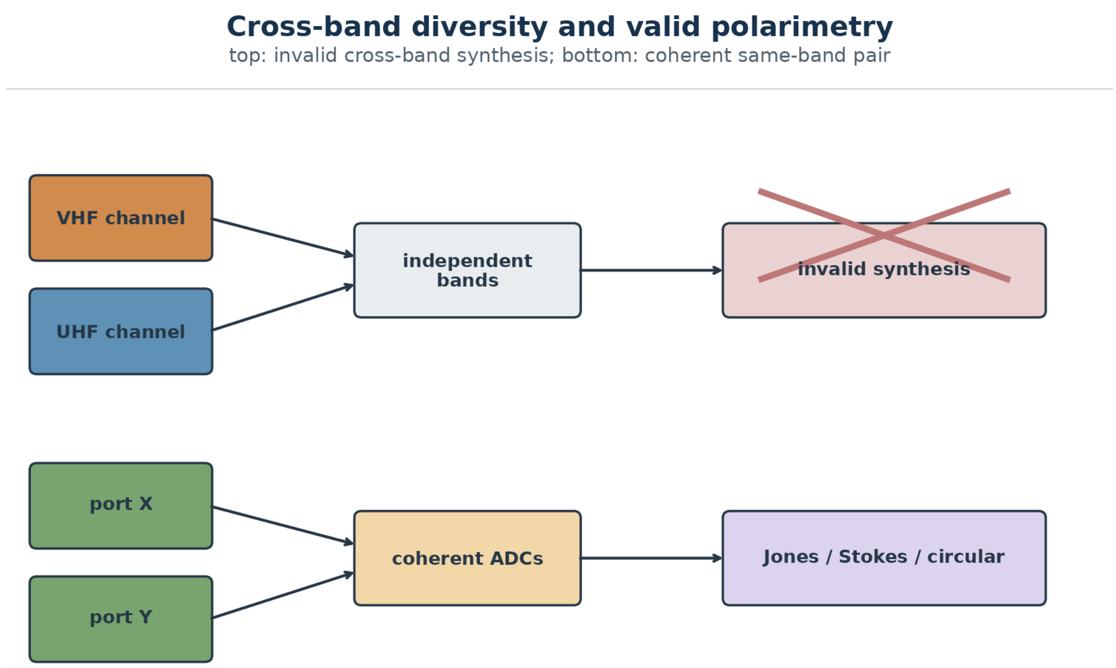

Figure 4.2: Top flow: independently digitized VHF and UHF channels terminate in a crossed-out cross-band polarimetric result. Bottom flow: two coherent orthogonal ports in the same band feed calibrated Jones, Stokes or circular-polarization products.

Chapter 5

Distributed Electronics and Event Processing

Each band has a specific antenna, preselector, limiter, LNA, converter when needed, coherent ADC, FPGA/ASIC and circular memory. The local chain is:

$$\text{antenna} \rightarrow \text{RF front-end} \rightarrow \text{ADC} \rightarrow \text{DDC/PFB/FFT} \rightarrow \text{detector} \rightarrow \text{event.}$$

The system performs three reductions: band-level detection, face-level fusion and global radome fusion. Raw samples remain local and are preserved around triggers. Normal network traffic consists of event descriptors, telemetry, summaries and selected I/Q captures rather than permanent aggregate ADC streaming.

For C3 storage screening, the aviation paths use a proposed 8 MS/s complex capture with 16-bit I and 16-bit Q for each of the simultaneous UAT and 1090ES channels (`RF-012`). A selected direct UHF waveform uses a proposed 25 MS/s complex channel (`RF-013`), while the bistatic protocol uses two coherent 25 MS/s channels for reference and surveillance (`RF-014`). These rates produce 512 Mbit/s, 800 Mbit/s and 1.6 Gbit/s per node respectively before framing or compression. Ten-second raw event windows therefore require 0.64 GB, 1.00 GB and 2.00 GB per node. They are capacity baselines, not claims that each waveform occupies the full sampled bandwidth.

Noise figure, IP3 and usable dynamic range remain deliberately unassigned (`RF-015`) until a measured site-RFI distribution, antenna gain pattern, preselector rejection and cascaded component budget exist. Component selection must close sensitivity and blocker survival together; an isolated low-noise figure is not sufficient.

Every antenna port assigned to a band has its own shielded acquisition module containing at least one ADC and one band ASIC. A dual-polarized aperture therefore uses two independent enclosures, one for each orthogonal coherent channel (`MEC-004`); sharing the sampling reference does not mean sharing an uncontrolled RF/digital cavity. The modules are separated by conductive partitions, RF gaskets and controlled cable/fibre feedthroughs. A central face module, the FFASIC, receives events from all band modules, performs intra-face correlation and interfaces with timing, thermal monitoring, power and the optical link.

Thermally, each shielded module is mounted to a conductive cold plate connected to a controlled heat path. The RF cavity, digital processing electronics and power conversion are physically partitioned so that heat sources and digital EMI do not share an uncontrolled volume with the sensitive RF aperture. Temperature sensors are attached to the LNA, ADC, ASIC, clock and cold plate; thermal drift is recorded as metadata and triggers recalibration when the validated range is exceeded.

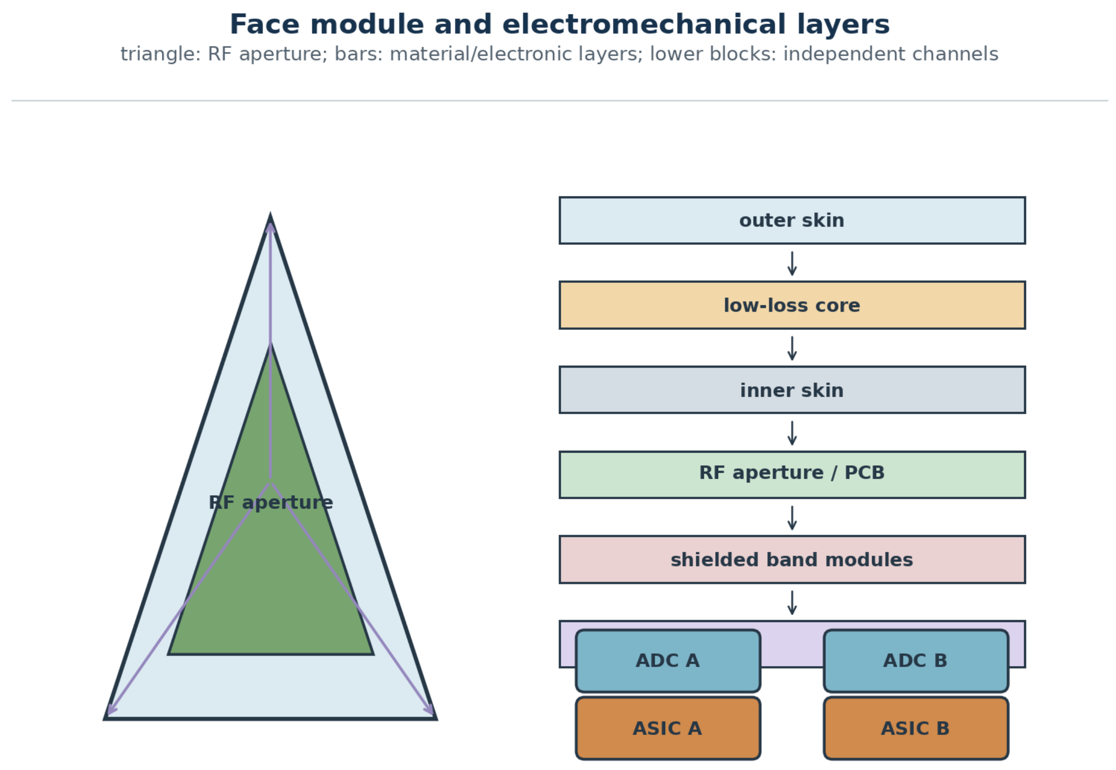

Figure 5.1: Triangular RF face module and exploded stack: the left triangle represents aperture, calibration and structural paths; the right coloured plates represent successive RF, dielectric, shielding, processing, thermal and service layers.

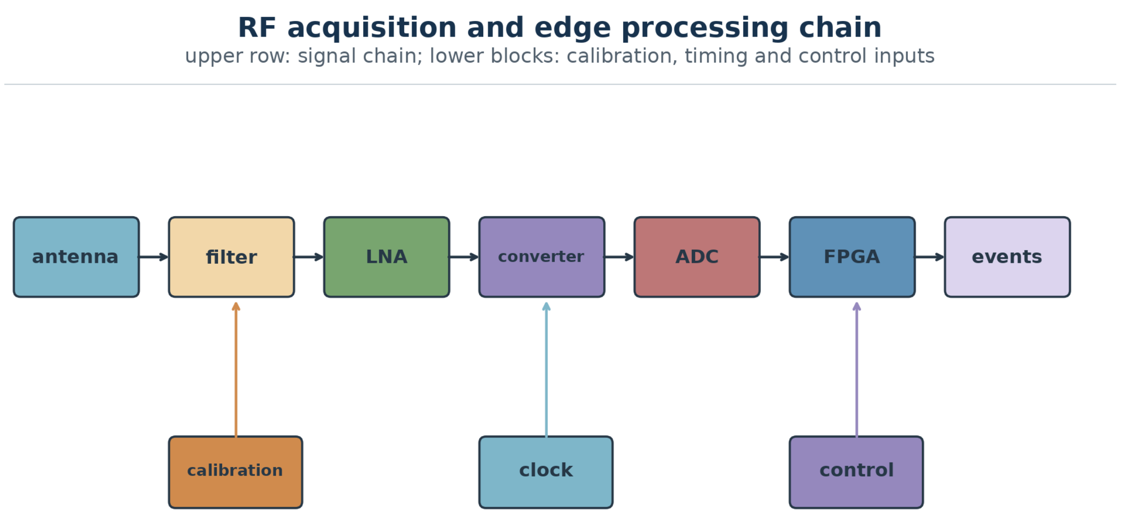

Figure 5.2: RF and edge-processing chain. The upper sequence represents antenna-to-event processing; the three lower blocks and vertical arrows represent calibration injection, common timing and supervisory telemetry.

Chapter 6

Timing, Calibration and Localization

White Rabbit or IEEE 1588 HA can distribute deterministic time and frequency over fibre, but timing synchronization is not automatically carrier-phase coherence at Ka. The phase uncertainty is:

$$\sigma_\phi = 2\pi f \sigma_t.$$

Every sample must be linked to a common epoch and corrected for measured channel delay:

$$t_{i,j,n} = T_{\text{epoch}} + \frac{n}{f_{s,i,j}} + \delta_{i,j}, \qquad t_i^* = t_i - \tau_i(f, T).$$

For an illuminator T, target x and receiver R_i, the bistatic excess path is:

$$\Delta\rho_i = \|x - T\| + \|x - R_i\| - \|T - R_i\| = c\Delta\tau_i.$$

AOA, TDOA and FDOA are fused with covariance, propagation models and geometric dilution of precision.

All faces and receiver chains within a radome share the node's central time base. Across radomes, that base is referenced to the network common epoch: GNSS and a frequency reference discipline the local clock, White Rabbit distributes time and frequency, atomic holdover preserves continuity, and calibration injection measures RF and digital delays. These techniques operate simultaneously and are checked continuously; each observation timestamp carries its calibration version and uncertainty. Echoes associated across several antennas of one radome and across other radomes can therefore be reconciled on the same time axis.

6.1 GNSS timing reference

Each station provisionally includes at least two external, surveyed GNSS antennas (`TIM-001`) with clear sky view, separated from the VHF/UHF Yagis and other high-power RF sources. A multi-constellation receiver produces UTC, 1PPS, frequency reference and ephemeris data. The proposed 1PPS and 10 MHz outputs (`TIM-002`) discipline a local atomic clock or high-stability oscillator; the clock then distributes deterministic timing through White Rabbit to every face and ADC.

The GNSS antenna phase centre, cable delay, receiver delay and distribution delay are measured and versioned. GNSS provides absolute epoch and long-term frequency discipline, while a local atomic holdover preserves continuity during loss of GNSS. End-to-end RF calibration remains mandatory:

$$t_i^* = t_i - \tau_{\text{GNSS},i} - \tau_{\text{cable},i} - \tau_{\text{RF},i} - \tau_{\text{ADC},i}.$$

Known station coordinates allow consistency checks between surveyed position, satellite geometry and measured network timing, but do not by themselves eliminate multipath, antenna phase-centre error or analog delay.

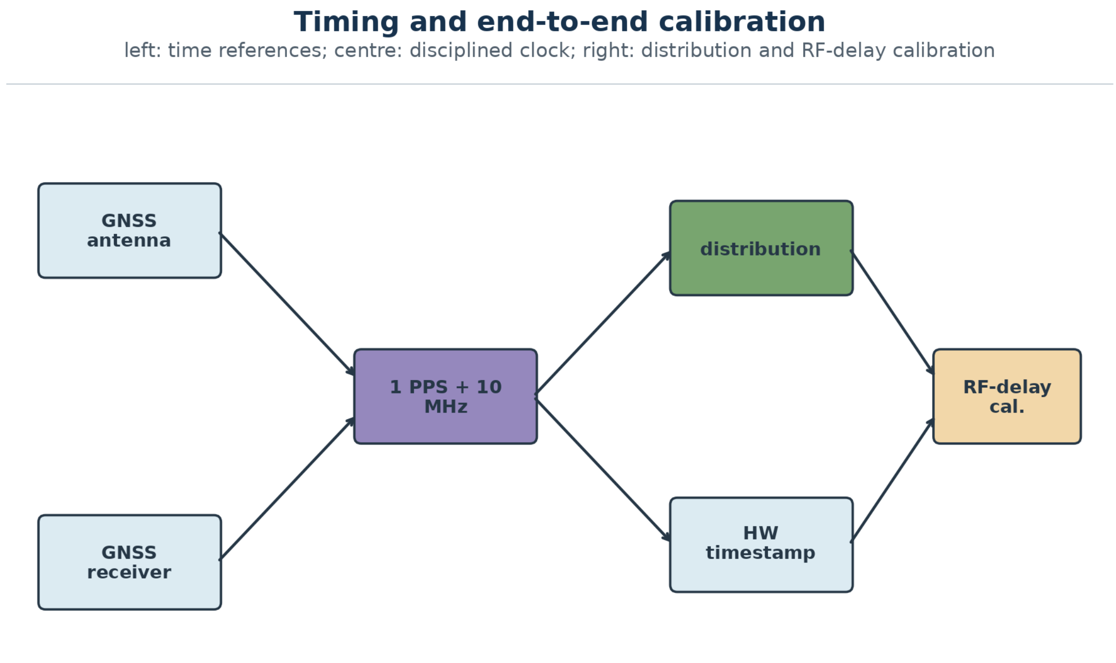

Figure 6.1: Timing and calibration architecture: the two left sources represent absolute GNSS reference and local holdover, the centre is the disciplined clock, the upper and lower right branches distribute timing to acquisition, and the final block represents measured RF-delay correction.

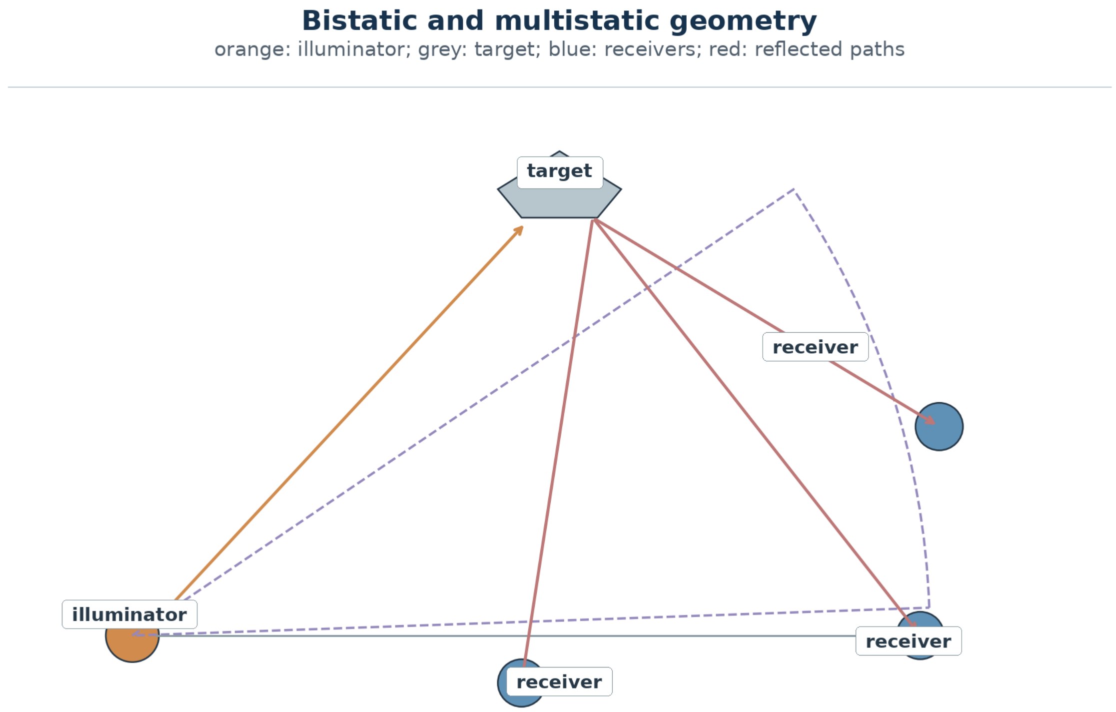

Figure 6.2: Bistatic/multistatic geometry: the orange point is the illuminator, the upper polygon is the target, blue points are receivers, orange and red paths are illumination and target-to-receiver propagation, and the dashed arc represents an equal-delay locus.

Chapter 7

Passive Radar Processing

The passive-radar chain equalizes reference and surveillance channels, cancels direct signal and clutter, computes the cross-ambiguity function, detects with false-alarm control and associates detections across nodes. The output is a track with uncertainty, not only a coordinate. Direct-emitter detection must remain distinct from bistatic-reflection detection. An earlier multiband demonstrator used four LPDAs, reception from 200 MHz to 2 GHz, simultaneous four-channel digitization and cross-ambiguity processing to exploit analogue and digital transmissions of opportunity [5]. This evidence supports the historical feasibility of the chain but does not replace RADOME validation.

7.1 Aircraft broadcast consistency experiment

The reference aircraft signal is ADS-B, primarily 1090ES at 1090 MHz (`EXP-001`) for commercial aviation. The alternative 978 MHz UAT channel (`EXP-002`) is contextual and relevant mainly to certain general-aviation deployments in the United States. Both are assigned to the dedicated 960–1215 MHz aviation aperture or array rather than to the 470–860 MHz UHF Yagi. A diplexed receiver provides simultaneous independent service paths, each with its own preselector, limiter/LNA, coherent ADC and FPGA channel. ADS-B messages are transmitted automatically and repeatedly, with message timing dependent on the information type; they are not one continuous packet carrying every parameter at every instant. Expected fields include GNSS-derived position, altitude, velocity, track and aircraft identification.

Two radome nodes (`EXP-005`) separated by a proposed 100 km baseline (`EXP-003`) receive the aircraft transmission independently. This illustrative geometry is distinct from the three-node demonstrator target (`ARC-002`). For each node, the system records received power P_i, local angle of arrival AOA_i, hardware timestamp t_i and Doppler estimate $f_{D,i}$. The fusion layer computes:

$$\Delta t = t_A - t_B, \qquad \Delta f_D = f_{D,A} - f_{D,B}.$$

Together with surveyed node coordinates, aircraft ephemeris hypotheses and propagation uncertainty, these observables produce a consistency region for altitude, position and velocity. The result checks whether the broadcast parameters are compatible with the physical observation; it is not a blind guarantee that every transmitted parameter is truthful.

ADS-B supplies a cooperative identity and state report, while the two-node RF geometry supplies an independent consistency check based on received power, AOA, TDOA and Doppler.

The nominal 100 km baseline remains a screening geometry, not an approved deployment distance. With the proposed screening heights of 1000 m for a station and 10000 m for an aircraft (`EXP-010`), a standard effective-Earth radio-horizon calculation gives approximately 542 km of mutual geometric horizon. This only shows that 100 km is not excluded by curvature in that example. Terrain masking, antenna patterns, link margin, common visibility, GDOP, spectrum authorization and actual site coordinates must still close before field deployment.

7.2 Independent UHF illuminators

Known UHF television or other broadcast transmitters can provide an independent triangulation test alongside ADS-B. Their direct signals can be used to calibrate AOA, timing, received-power mapping and site geometry when transmitter coordinates and waveform characteristics are known. Their reflected signals can also support a separate bistatic experiment, but direct-emitter verification and target-reflection localization must not be mixed in the same estimator without an explicit propagation model.

The experiment should compare: (i) ADS-B at 1090 MHz as a cooperative direct emitter; (ii) known UHF broadcast signals, provisionally in the 470–860 MHz region (`EXP-004`), as direct reference illuminators; and (iii) bistatic echoes from a controlled or independently tracked target. Agreement across these channels tests the radome network's triangulation, calibration and covariance reporting rather than merely repeating the aircraft message.

7.3 Independent experiment protocols

The cooperative direct-emitter protocol (`EXP-006`) records decoded identity and state, received power, AOA, hardware time and Doppler. Its truth source is the decoded GNSS-derived state with independently surveyed receiver coordinates; its estimator tests RF-versus-message consistency, and its metrics are association rate and position, velocity, AOA, time and Doppler residuals with declared uncertainty.

The known-transmitter calibration protocol (`EXP-007`) records the direct UHF waveform, received power, AOA and calibrated channel delay. Its truth source is an independently surveyed transmitter position, antenna sector and waveform metadata; its estimator solves the expected direct path, and its metrics are AOA, delay and power-map residuals plus repeatability over time and temperature.

The bistatic-reflection protocol (`EXP-008`) uses simultaneous reference and surveillance channels, direct-path cancellation and a cross-ambiguity estimator. Its truth source is a controlled or independently tracked target trajectory plus surveyed transmitter and receiver coordinates; its metrics are probability of detection, false-alarm probability and delay, Doppler and localization errors with covariance consistency. No result from the two direct-signal protocols substitutes for this bistatic evidence.

7.4 Bistatic ellipsoid and uncertainty-region reduction

Let the illuminator position be $\mathbf{s}$, the surveyed position of receiver i be $\mathbf{r}_i$, and a hypothesized target position be $\mathbf{x}$. The excess delay between the echo and the direct reference path satisfies

$$c\,\Delta\tau_i = \|\mathbf{s}-\mathbf{x}\| + \|\mathbf{x}-\mathbf{r}_i\| - \|\mathbf{s}-\mathbf{r}_i\|.$$

For fixed $\Delta\tau_i$, compatible points form a bistatic iso-range ellipsoid whose foci are the transmitter and receiver. Per-face received power, corrected for the embedded pattern, losses and the illuminator's effective radiated power, supplies an angular/link likelihood; AOA restricts the solution to a direction or cone. Polarization observables add discrimination only at the same frequency and through calibrated coherent ports. The present VHF and UHF Yagis operate in different bands and therefore cannot be crossed to form a polarimetric measurement of the same echo.

The probabilistic intersection of delay ellipsoids, AOA, Doppler/FDOA, power and valid polarization yields a state region with covariance rather than an exact point. Associated reception of the same echo by multiple antennas and faces of one radome adds angular, power, delay and Doppler observations to the estimator. Reception of that echo by a second synchronized radome adds another ellipsoid and further constraints; additional integrated nodes add independent information and help reject ambiguities.

In weighted-least-squares reconciliation, or an equivalent probabilistic estimator, every additional valid observation adds a positive-semidefinite information matrix. The estimated covariance therefore cannot increase, and measurement uncertainty decreases strictly along directions for which the new observation supplies non-redundant information. This joint propagation of input quantities, covariances and multivariate output quantities follows the metrological treatment in GUM Supplement 2, JCGM 102:2011 [7]. It is a mathematical consequence of the fusion model, not a claim of already measured performance. Prior knowledge of the illuminator position and waveform, reference channels, hardware timestamping, and the common time base—centrally referenced and continuously calibrated through GNSS, frequency reference, White Rabbit, holdover and delay measurement—allow the observations to be associated and reconciled. The demonstrator must quantify the magnitude of the reduction and compare predicted covariance with observed error and GDOP; geometry, SNR, multipath, calibration and error correlation determine the achieved gain, not the existence of the reduction principle.

7.5 Jamming as an observable emission

Electronic attacks against active radars may use noise, sweep, frequency agility, and coherent or deceptive waveforms. To RADOME, this energy does not disappear: it is an additional direct emission in the spectrum and increases passive detectability of the transmitter when it lies within the covered bands and dynamic range, especially at high power, bandwidth or duty cycle. NATO literature treats passive radar, passive emitter localization and coherent electronic attack against sensors—including PCL—as related but distinct problems [11, 12].

Protocol `EXP-011` treats jamming as a non-cooperative source. Multiple faces and radomes record power, spectrum/occupancy, AOA, timestamps and FDOA/Doppler; temporal and spectral association support estimation of the emission centroid and, through a sequence of states, its motion vector. Increasing radiated energy to obscure active radars therefore supplies the passive network with more direct energy to intercept and geolocate. Processing must preserve raw data before interference suppression, separate jammer detection/classification from the bistatic-echo chain, and use limiters, preselection and gain control to prevent saturation.

This advantage does not automatically identify the aircraft. Directional, low-probability-of-intercept, coherent, towed, expendable or stand-off jammers can create multiple sources, false angles and ambiguous association; the localized emitter may not be aboard the fighter. The controlled claim is that jamming increases the electromagnetic observability of its source to the passive network and creates additional observables. Tests must measure probability of intercept, AOA/TDOA/FDOA error, saturation, classification and track consistency for each signal class.

7.6 Cellular towers as calibrated illuminators

Cellular base stations add a particularly valuable set of known illuminators. Their surveyed coordinates, sector orientations and nominal carrier allocations allow the network to use multiple cellular frequencies and waveforms as direct references. The direct signal can calibrate coverage, AOA, timing and received-power consistency; reflections from aircraft, terrain or controlled targets can then test bistatic detection across several carriers and bandwidths. Each carrier must be treated as a separate waveform and propagation case, with transmitter power, antenna sector, scheduling, synchronization and local RFI recorded as metadata.

This converts the cellular infrastructure into a distributed test field for the wideband multi-radome radar concept. It does not turn the system into an active radar: the radomes remain passive receivers, and the useful observables come from known transmissions, their geometry and their reflected components.

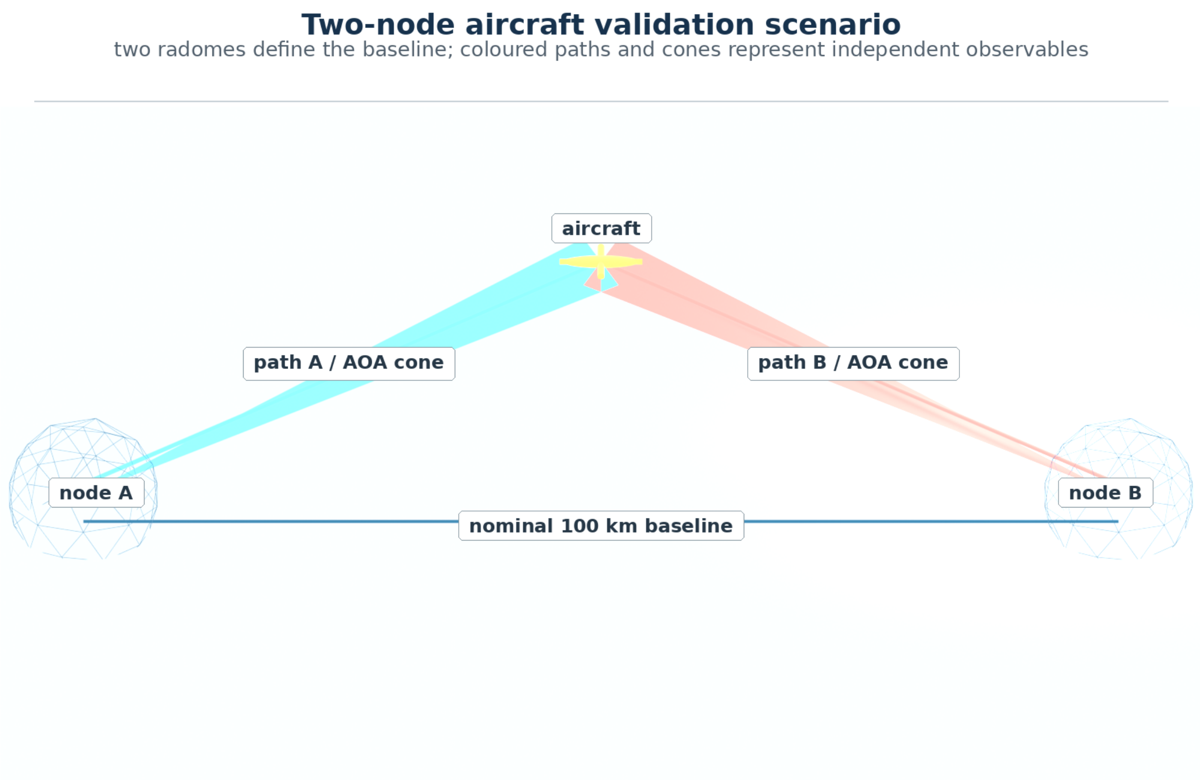

Figure 7.1: Two-node aircraft scenario: the two enlarged radomes define the nominal 100 km baseline, the aircraft is above and offset from it, and the blue/red paths and cones represent independent received signals and angular observations used for power, AOA, TDOA and Doppler consistency.

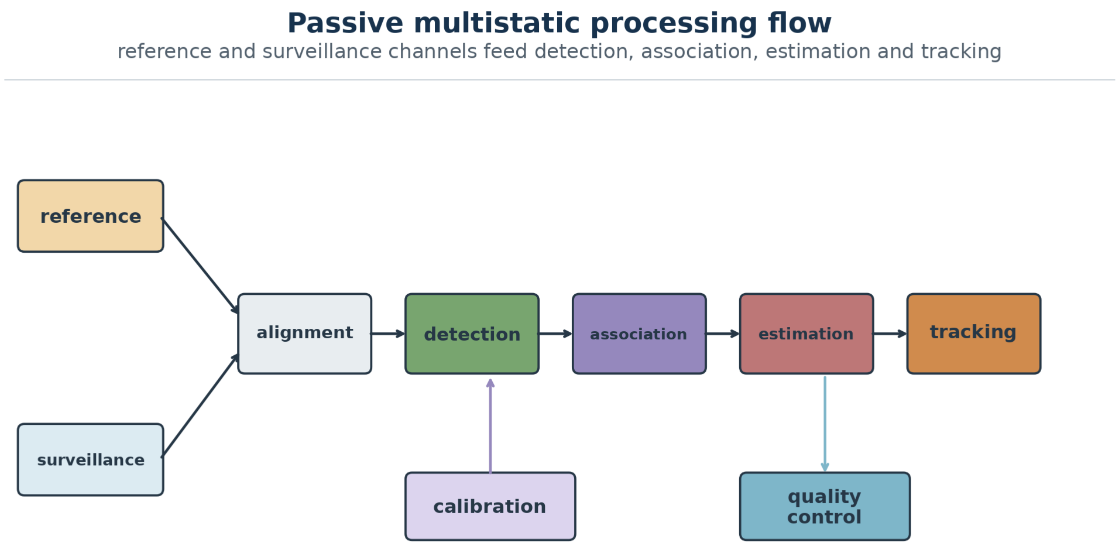

Figure 7.2: Multistatic passive-processing flow: reference and surveillance inputs converge into channel equalization, direct-path cancellation, delay–Doppler processing, controlled detection and association; lower branches represent emitter priors and the resulting state estimate.

Chapter 8

Mountain Infrastructure, Validation and Risks

The station requires resilient power, UPS, batteries, generator, optical links, thermal management, lightning protection, EMC control, perimeter security and logistics. Site selection must consider terrain, horizon, local RFI, climate, access, energy and fibre availability. Environmental qualification must cover wind, vibration, rain, humidity, salinity, UV and thermal gradients.

The threat model assumes that every site will eventually be located, photographed and mapped by potential rivals possessing optical or synthetic-aperture-radar remote sensing. SAR systems form images of fixed structures from microwave echoes and operate independently of sunlight and, to a large extent, cloud cover [4]. Security and operational continuity therefore cannot depend on keeping coordinates secret, on the dome's visual appearance or on an expectation that its echo will never be observed. The architecture assumes prior adversary cataloguing and instead relies on absence of self-generated emissions, geographic distribution, inter-node redundancy, modular maintenance, physical/cyber protection and recovery after loss of a station. A low emission signature remains valuable because it hinders inference of activity, operating mode and collected data even when the physical location is already known.

The timing infrastructure includes at least two external GNSS antennas with surveyed coordinates and phase centres. They are mounted above or near the radome with clear sky visibility, physical separation from the VHF/UHF Yagis and filtered, shielded feeds to the timing room. Redundant multi-constellation receivers support cross-checking, fault detection and atomic-clock disciplining.

The proposed civil interface is a reinforced-concrete parallelepiped base measuring approximately $4\,\mathrm{m} \times 4\,\mathrm{m}$ in plan (`CIV-001`) and $3\,\mathrm{m}$ in clear structural height (`CIV-002`). Its upper slab is intended to interface with the proposed lower radome cut (`GEO-007`); the support geometry remains blocked until C2 verifies the required transition or base size. A service opening is provided in the front wall so personnel and equipment can enter the interior without removing RF panels.

The thermal solution is zoned. Heat generated by LNAs, ADCs, ASICs, FPGAs, optical transceivers and DC/DC converters is conducted to local cold plates and rejected through monitored air or liquid paths. The central timing and fusion core has an independent thermal zone. Condensation sensors, inlet/outlet temperature sensors and power telemetry support closed-loop control. Thermal phase drift is measured, timestamped and included in the calibration model.

The structural solution uses an internal load-bearing frame with sealed joints, node plates and gussets. Each external Yagi support/boom axis continues from the internal pyramid apex through the triangular face, while dielectric panels provide environmental protection and controlled RF transmission. Acceptance therefore covers static strength and dynamic stability: wind, vibration, modal response, thermal expansion, rain loading, maintenance access and repeated thermal cycles.

8.1 Cost–performance and maintainability hypothesis

Face-level modularity allows one receiver chain to be tested, calibrated, removed and replaced without disassembling an electromagnetically transparent ultra-wideband enclosure. The absence of a dedicated transmitter also removes the node's power amplifier, high-power duplexing and part of the cooling and safety burden associated with active radar. Conversely, exposed antennas increase wind area, corrosion, icing, lightning, mechanical-damage and feedthrough-sealing risks and may require more frequent inspection. The architecture is economically preferable only if avoided shell/transmitter costs exceed these added costs and the cost of multiplying faces and channels.

A controlled comparison must record bill of materials, tooling, fabrication/integration labour, production yield, calibration instrumentation, energy, spares, mean time to repair, downtime and site maintenance over a common reference life. Performance must be normalized by spectral/angular coverage, realized gain, sensitivity, localization accuracy/covariance, availability and resilience. Without that model and traceable quotations, the literature supports greater material and manufacturing complexity in multilayer UWB radomes but does not prove lower RADOME total cost.

The development path is P0 system simulation, P1 three-node VHF/UHF demonstrator, P2 L/S/C tile, P3 X/Ku/Ka tile and P4 integrated outdoor campaign. Acceptance tests include S-parameters, 3D OTA patterns, radome insertion loss and phase, coherence, synchronization, calibrated AOA, passive-radar Pd/Pfa and resilience after link or power loss.

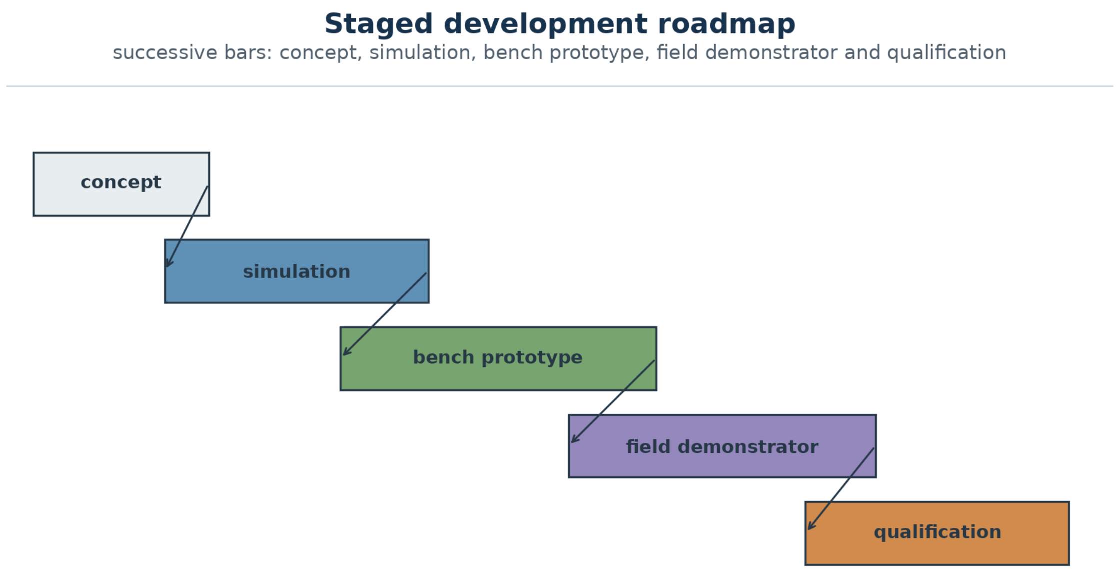

Figure 8.1: Development and validation roadmap: the five staggered coloured bars represent successive, partly overlapping gates from requirements and the VHF/UHF demonstrator through higher-band tiles and the integrated environmental campaign.

Chapter 9

Literature Review, Innovation and Gaps

The literature is organized into three relatively mature but weakly integrated streams: conformal broadband antennas and arrays; functional radomes and frequency-selective surfaces; and passive detection, localization and emitter identification. Conformal direction-finding studies show that mutual coupling, supporting structure and radome distortion belong to the calibrated array manifold, not merely to a mechanical enclosure [2, 8].

In engineering literature, *radome* contracts *radar dome* and denotes the structure that protects an antenna from the environment under a trade-off between structural integrity and electromagnetic transparency [16]. The paper supplied in `bibliography/` evaluates this trade-off for active radar and communication systems and shows that transmission loss, oblique incidence, co-polar mismatch and depolarization must be measured with the antenna and radome integrated. The structural definition does not, however, require the protected antenna to transmit. The present architecture specializes the concept as a receiving dome: the node has no dedicated RF illuminator, and every surveillance chain is receive-only.

Functional-radome research demonstrates passband transmission combined with absorption or diffusion outside selected bands [3, 10, 14, 18]. This creates a design tension for open spectrum monitoring: a radome optimized for stealth or narrow transmission can reject signals that a surveillance receiver should observe.

The newly supplied local documents extend this stream in two directions. Ramanamurthy and Krishna describe the mechanical/electromagnetic co-design of a composite high-pressure radome with claimed coverage from 15 kHz to 18 GHz; that extreme value is not transferred to this project without independent reproduction, but the work reinforces that material, thickness, joints, moisture uptake, structural load and insertion loss must be verified on the same prototype [17]. More directly relevant, Oh et al. demonstrate a 338–470 MHz UHF package in which the antenna, metallic isolation wall and sandwich GFRP radome are integrated from the outset. Their measured prototype achieved 32.7% impedance bandwidth and less than 0.4 dB radome insertion loss with little pattern distortion [13]. Those values characterize that airborne package and are not assumed targets for RADOME.

The theses by Abotalebi and Tandel document compact alternatives to Yagis, including monocones, helix-fed bicones, multilayer/FSS patches, cavities and dielectric covers [1, 21]. Their results confirm the trade-offs among electrical size, bandwidth, gain, efficiency, polarization, manufacturing and cover interaction. They justify retaining an experimental topology comparison in the roadmap, but do not demonstrate that one antenna provides uniform performance across every proposed band. The locally available Tandel source is only a preview, so it supports the stated design scope rather than detailed performance values.

Passive-localization research uses AOA, TDOA and FDOA, with accuracy governed by geometry, synchronization, propagation and computational load [9, 15]. Gould et al. document a low-cost multiband PCL demonstrator assembled from commercial components, capable of exploiting analogue and digital broadcasts and producing DVB-T aircraft detections consistent with ADS-B data [5]. The work identifies advantages over active radar from using third-party transmitters: procurement and operating-cost savings, lower power demand and covertness through the absence of dedicated illumination; observations across distinct bands and geometries also provide diversity for more robust and accurate track fusion. Those benefits carry explicit limits: the prototype required more than 90–100 dB of dynamic range, inter-channel amplitude/phase calibration and direct-path suppression, while the achieved digital cancellation was still insufficient for reliable detection. The source therefore supports the architectural advantages of passive sensing while reinforcing RADOME's dynamic-range, calibration and cancellation gates, without transferring its experimental results to this project. RF fingerprinting and open-set anomaly detection can assist emitter characterization, but closed-set classifiers and limited training conditions cannot be treated as universal solutions [6, 19, 20].

This separation supports two passive inference classes that must not be conflated. Direct signals from an emitter may be located by AOA triangulation or TDOA/FDOA multilateration. For non-emitting targets, the network uses external illuminators of opportunity and compares a direct reference channel with surveillance channels containing bistatic or multistatic reflections. In both cases the radomes only receive; position follows from geometry and calibrated multi-node fusion rather than from an interrogation transmission generated by the system.

The project gap is the end-to-end integration of a conformal multiband aperture, distributed passive reception, polarimetric and temporal calibration, hierarchical event processing and realistic mountain-site validation. The innovation is therefore systemic and must be demonstrated with reproducible measurements.

9.1 Candidate architectural innovation: external aperture and absorbing cell

The proposed combination of VHF/UHF Yagis exposed beyond the shell with individually shielded, absorber-lined internal pyramidal cells defines an integration gap to be investigated. The external antenna keeps the useful path from traversing the dielectric cover; behind each face, the cell is intended to confine sensitive electronics, limit inter-channel coupling and dissipate the fraction of energy that enters the aperture without being delivered to the receiver. In principle, this may combine efficient reception of existing illuminators, lower self-interference and lower reradiation from the cavity. Absorptive and diffusive radome literature shows that selective transmission, absorption and scattering control can be co-designed [3, 10]; Oh et al. already demonstrate that a metallic wall, UHF antenna and protective radome can be integrated and measured as one package [13]. None of these sources, however, validates the specific combination of two external orthogonal different-band Yagis, a face-normal boom and an absorber-lined internal pyramidal cell in a distributed receive-only network.

Two signatures must be distinguished. The emission signature is necessarily lower than that of an equivalent active radar because RADOME generates neither illumination nor RF scanning: it only receives known external emissions and their echoes. To an external sensor, an electromagnetic contribution caused by the station's presence is secondary scattering of ambient illumination rather than a transmission originated by the system. This low detectability through emissions is a property of the passive architecture, not a performance claim awaiting demonstration.

The structure's scattering signature is a separate question. A conductive Faraday wall without dissipative material reflects the incident wave; moreover, the exposed Yagis, boom, edges, seams and cables continue to scatter energy. The additional low-observability magnitude obtained from the absorbing cell requires selecting its material and thickness and measuring, versus frequency, polarization and angle, the balance among received, reflected, absorbed and reradiated power, as well as aperture S_{11}, shielding effectiveness, inter-cell coupling and monostatic/bistatic RCS with and without the lining. The paper therefore asserts emission discretion while retaining scattering-signature reduction as a measurable hypothesis; it does not claim absolute invisibility.

9.2 Gap coverage by the present architecture

The current design resolves several gaps at the architectural level. The calibrated manifold includes the radome, face geometry, mutual coupling and local antenna orientation; the hybrid aperture assigns one external single-polarization VHF Yagi and one smaller single-polarization UHF Yagi to independent RF modules; same-band dual-port modules are required wherever polarimetric synthesis is claimed; each antenna has a shielded ADC/ASIC chain; the FFASIC performs face-level fusion; and the thermal and structural design defines cold plates, sensors, internal load paths and environmental gates. These are design resolutions, not measured performance claims. The remaining work is to quantify insertion loss, phase and group delay, structural margins, thermal gradients, EMC isolation and localization covariance in representative tests.

9.3 Gap coverage by the band-separated aperture

The review identified a missing end-to-end treatment of aperture integration, polarization, calibration and realistic platform effects. The present architecture addresses this at the design level by separating cross-band mechanical integration from same-band polarimetry. The VHF and UHF Yagis provide independent spectral observations; only a dual-port module within one band may produce calibrated Jones, Stokes, RHCP or LHCP quantities. The remaining research question is quantitative: measured gain, bandwidth, isolation, cross-polarization for valid dual-port modules, platform effect and stability must confirm the design.

Chapter 10

Conclusion and Next Steps

The consolidated project defines a technically defensible research platform: distributed geodesic radomes, a hybrid aperture with external VHF Yagi antennas and smaller higher-frequency elements, multiband vector reception, local event processing, precise timing, end-to-end calibration and multistatic fusion. The central design principle is not a universal antenna, but a mechanically integrated family of antennas sharing a common face-level timing, calibration and fusion interface.

As a candidate architectural innovation, the project combines the external Yagis with individually shielded receiving pyramidal cells whose absorber lining remains to be selected. Receive-only operation already establishes a low emission signature relative to active radar: there is no self-generated illumination, and any return caused by the station is only secondary scattering of external waves. The additional hypothesis is to preserve the receive path outside the shell while reducing self-interference, coupling, aperture reradiation and scattering signature. The magnitude of this additional gain must be demonstrated through power balance, shielding effectiveness, OTA patterns and angular RCS; the document does not claim absolute invisibility or operational performance.

The crossed-band Yagi assembly is a proposed mechanical integration solution, not yet a demonstrated innovation. A larger external VHF Yagi and a smaller orthogonally mounted UHF Yagi combine distinct spectral roles in one face-level interface, but they remain independent single-polarization channels and are not combined for polarimetric synthesis. Full polarimetry is reserved for coherent dual-port modules within the same band.

The lowest-risk next step is a three-node VHF/UHF demonstrator with reference and surveillance channels, followed by L/S/C and X/Ku/Ka tiles. HF should proceed as a dedicated program for vector sensors and characteristic platform modes. Territorial expansion must wait for measured coverage, link budgets, geometry, false-alarm performance and metrological stability.

Appendix A

Minimum Detection Record

Field	Description
station_id	Node identifier and calibration version
tile_id/channel_id	Tile, polarization port and RF chain
timestamp_utc	Time with estimated uncertainty
frequency_hz/bandwidth_hz	Observed centre frequency and bandwidth
polarization	Measured port/orientation for single-polarization channels; Jones or Stokes only for a calibrated coherent same-band pair
aoa_az_el	Local direction and covariance
delay_doppler	Bistatic measurements and quality
illuminator_id	Illuminator identity or hypothesis
sync_state	locked, holdover or degraded
environment	Temperature, humidity and radome state

Bibliography

[1] Shayan Abotalebi. Ultra-wideband uhf/vhf electrically small antennas and ultra-wideband s-band and c-band compact form-factor antennas. Master's thesis, New Mexico State University, Las Cruces, New Mexico, 2023. ProQuest 30695987; primary thesis supplied locally in bibliography/.

[2] D. Caratelli, I. Liberal, and A. Yarovoy. Design and full-wave analysis of conformal ultra-wideband radio direction finders. *IET Microwaves, Antennas & Propagation*, 5: 1164–1174, 2011. URL https://consensus.app/papers/details/5f166a5e2c715f25a2cfa7d55f20749f/?utm_source=chatgpt. Consensus citation count reported in this review session: 17.

[3] F. Costa and A. Monorchio. A frequency selective radome with wideband absorbing properties. *IEEE Transactions on Antennas and Propagation*, 60:2740–2747, 2012. URL https://consensus.app/papers/details/1811b78b36fa5cc0840c16f9dd6f2a1b/?utm_source=chatgpt. Consensus citation count reported in this review session: 678.

[4] European Space Agency. Introduction to a sar system. ESA Space Engineering and Technology, n.d. URL https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Onboard_Data_Processing/Introduction_to_a_SAR_System. Accessed 9 August 2026.

[5] Dale Gould, Robert Pollard, Carlos Sarno, and Paul Tittensor. A multiband passive radar demonstrator. In *2006 International Radar Symposium*. IEEE, 2006. doi: 10.1109/IRS.2006.4338144. URL https://doi.org/10.1109/IRS.2006.4338144.

[6] Anu Jagannath, Jithin Jagannath, and P. Kumar. A comprehensive survey on radio frequency (rf) fingerprinting: Traditional approaches, deep learning, and open challenges. *Computer Networks*, 219:109455, 2022. URL https://consensus.app/papers/details/85a1acc05180584992b4bd752a62599e/?utm_source=chatgpt. Consensus citation count reported in this review session: 263.

[7] Joint Committee for Guides in Metrology. Evaluation of measurement data—supplement 2 to the guide to the expression of uncertainty in measurement—extension to any number of output quantities. Technical Report JCGM 102:2011, Bureau International des Poids et Mesures, 2011. URL https://doi.org/10.59161/JCGM102-2011.

[8] I. Liberal, D. Caratelli, and A. Yarovoy. Conformal antenna array for ultra-wideband direction-of-arrival estimation. *International Journal of Microwave and Wireless Technologies*, 3:439–450, 2011. URL https://consensus.app/papers/details/962901757e715c669ba9962b91c0b96b/?utm_source=chatgpt. Consensus citation count reported in this review session: 9.

[9] Zhixin Liu, Rui Wang, and Yongjun Zhao. Computationally efficient tdoa and fdoa estimation algorithm in passive emitter localisation. *IET Radar, Sonar & Navigation*, 2019. URL https://consensus.app/papers/details/47ae35140d8e563486d4808c6acd8485/?utm_source=chatgpt. Consensus citation count reported in this review session: 14.

[10] Qihao Lv, C. Jin, Binchao Zhang, and Zhongxiang Shen. Hybrid absorptive-diffusive frequency selective radome. *IEEE Transactions on Antennas and Propagation*, 69:3312–3321, 2021. URL https://consensus.app/papers/details/62fa6b42c0305286b6a3a4c60ea60a46/?utm_source=chatgpt. Consensus citation count reported in this review session: 67.

[11] NATO Science and Technology Organization. Coherent electronic attack on advanced radar systems (sci-252). Annual report, North Atlantic Treaty Organization, 2017. URL https://www.nato.int/nato_static_fl2014/assets/pdf/pdf_topics/20180522_STO_Annual_Report_2017.pdf.

[12] NATO Science and Technology Organization. Passive radar technology (set-243). Science and technology organization highlights, North Atlantic Treaty Organization, 2019. URL https://www.sto.nato.int/publications/Management%20Reports/2019_NATO_STO_Highlights_Web.pdf.

[13] Jaecheol Oh, Maengchang Kang, Junpyo Jo, Seungwoo Bang, Hyeon-Seok Choe, Sung-Hun Ha, Seokyoung Park, Jinbong Kim, Sangkeun Kim, Jungsuek Oh, and Hong-Kyu Jang. A compact broadband omnidirectional top-loaded uhf antenna with integrated ground wall and gfrp radome for conformal airborne applications. *Aerospace*, 13(3):227, 2026. doi: 10.3390/aerospace13030227. URL https://doi.org/10.3390/aerospace13030227. Primary source supplied locally in bibliography/.

[14] A. Omar and Zhongxiang Shen. Thin 3-d bandpass frequency-selective structure based on folded substrate for conformal radome applications. *IEEE Transactions on Antennas and Propagation*, 67:282–290, 2018. URL https://consensus.app/papers/details/905ccf75368255a78a236f083a31e9ee/?utm_source=chatgpt. Consensus citation count reported in this review session: 77.

[15] Karleigh Pine, Samuel Pine, and M. Cheney. The geometry of far-field passive source localization with tdoa and fdoa. *IEEE Transactions on Aerospace and Electronic Systems*, 57:3782–3790, 2021. URL https://consensus.app/papers/details/035e2d1ded6e5316b49b04c2007445cb/?utm_source=chatgpt. Consensus citation count reported in this review session: 36.

[16] Zeeshan Qamar, Jorge L. Salazar-Cerreno, and Nafati Aboserwal. An ultra-wide band radome for high-performance and dual-polarized radar and communication systems. *IEEE Access*, 8:199369–199384, 2020. doi: 10.1109/ACCESS.2020.3032881. URL https://doi.org/10.1109/ACCESS.2020.3032881. Primary source supplied locally in bibliography/.

[17] V. V. Ramanamurthy and L. Siva Rama Krishna. Design of high pressure radome with uwb frequency coverage for submarine communications. *International Journal of Mechanical Engineering and Technology*, 7(4):96–113, 2016. Primary source supplied locally in bibliography/; broad performance claims require independent reproduction.

[18] X. Sheng, Hongwei Wang, Ning Liu, and Kexin Wang. A conformal miniaturized bandpass frequency-selective surface with stable frequency response for radome applications. *IEEE Transactions on Antennas and Propagation*, 72:2423–2433, 2024. URL https://consensus.app/papers/details/6cb5b6043f755325b7abd988656c9afd/?utm_source=chatgpt. Consensus citation count reported in this review session: 53.

[19] Naeimeh Soltanieh, Y. Norouzi, Yang Yang, and N. Karmakar. A review of radio frequency fingerprinting techniques. *IEEE Journal of Radio Frequency Identification*, 4:222–233, 2020. URL https://consensus.app/papers/details/1c3cd42586205e0280356e90cb7ab147/?utm_source=chatgpt. Consensus citation count reported in this review session: 339.

[20] Liting Sun, Da Ke, Xiang Wang, Zhitao Huang, and Kaizhu Huang. Robustness of deep learning-based specific emitter identification under adversarial attacks. *Remote Sensing*, 14: 4996, 2022. URL https://consensus.app/papers/details/29ea0ff4b51857cca284e60d8c9c9d19/?utm_source=chatgpt. Consensus citation count reported in this review session: 22.

[21] Drasti K. Tandel. Design and development of an extremely wideband miniaturized monocone antenna with high gain attributes. Master's thesis, San Diego State University, 2024. Local preview supplied in bibliography/; used only for its stated design scope.